Certified Soft-Output List Evaluation for Coded SP-AFDM With Fractional Doppler

***Abstract*—Code-aided reception with superimposed pilots (SP) can benefit from examining multiple data–channel hypotheses, but evaluating a large codeword list under a full fractional-Doppler affine frequency division multiplexing (AFDM) model is expensive. Conventional hard pruning can preserve the list's maximum a posteriori (MAP) decision while discarding probability needed for soft decoding and channel estimation. This paper develops a certified computation-adaptive evaluator for a declared finite codeword list. Complete coded candidates are transformed to time-domain waveforms, yielding exact fractional-path regressors without truncating Doppler leakage. Gaussian gain marginalization is evaluated through normalized, nonnegative observation innovations. Their partial sums bound the unfinished likelihoods and support a probability-reserve pruning rule. The evaluator preserves finite-list MAP, deterministically bounds the discarded list probability, and controls the resulting posterior channel-mean change. Bit-probability and log-likelihood-ratio intervals are obtained from the same retained and bounded masses. Controlled full-propagation tests demonstrate reduced likelihood work, including under estimated geometry. Representative 1,024-candidate rescue-stage measurements show 34.7% and 21.9% time reductions at block lengths 96 and 192 relative to the same compiled recursion without pruning. A full-size frozen-receiver attachment verifies the interface but exposes dominant proposal cost; complete-receiver acceleration is not inferred from component timings. The guarantees concern the specified list and working channel model, not ungenerated codewords or unknown-support recovery.**

***Index Terms*—Affine frequency division multiplexing, superimposed pilots, soft-output list decoding, channel marginalization, adaptive computation, certified pruning.**

I. Introduction and Related Work

AFFINE frequency division multiplexing (AFDM) uses a chirp-parameterized transform to represent doubly selective channels and exploit delay–Doppler diversity [1]. Superimposed pilots (SP) avoid reserving pilot-only positions at the expense of coupling channel estimation to unknown data. Iterative SP-AFDM estimation and detection [2], basis-expansion/message-passing SP-OTFS receivers [3], and decoder-assisted interference cancellation [4] already establish the value of exploiting data information. The question addressed here is not whether joint reception is useful, but how to evaluate its competing code–channel explanations at controlled cost.

A practical receiver can propose a finite set of valid codewords using equalizer or decoder reliabilities. Evaluating more candidates can recover a word omitted by a short list. With uncertain channel gains, each candidate changes both the predicted SP waveform and the channel-marginalized likelihood. Fully evaluating every candidate can therefore become the dominant cost of a rescue stage. Fractional Doppler makes a dense transform-domain matrix an unattractive representation, although the underlying sampled path action remains structured.

Stopping a candidate once it cannot beat the incumbent is sufficient for hard decoding. It is not sufficient for soft output. Numerous nonwinning words can collectively carry appreciable probability; normalizing only the completed candidates then suppresses uncertainty and changes the posterior channel mean. The distinction is material in joint reception, where a distribution can be consumed by channel estimation after its maximizing word has already stabilized. A useful accelerator must therefore specify what soft information it preserves, not just which hard decision it returns.

A. Closest methods and the missing contract

Partial-cost search and recursive least squares are established. Al-Naffouri *et al.* [5] develop low-complexity blind OFDM equalization using partial-sequence criteria and recursive updates. Zaib and Al-Naffouri [6] study blind and semiblind branch–estimate–bound detection for space-time-block-coded OFDM. These provide the closest search ancestry. Their inspected criteria fit channel parameters while exploring transmitted-symbol hypotheses. The present evaluator instead integrates a specified gain prior, evaluates complete coded AFDM candidates through observation prefixes, and protects the soft distribution used by a subsequent channel-information consumer.

Soft-output search complexity is also not new. Feng *et al.* [7] improve GRAND, GCD, OSD, and SCL confidence by accounting for unvisited codewords under explicit bitwise-likelihood and codebook assumptions. Low-complexity soft-output OSD [8] and adaptive soft sphere-decoder clipping [9] trade computational effort for soft-output quality. Their confidence approximations or max-log clipping targets should not be identified with an exact bound on the sum of discarded weights under an arbitrary coupled, channel-marginalized likelihood.

Domain-specific structure is another established ingredient. Dual-domain AFDM detection [10] exploits time-domain sparsity, while data-aided fractional-parameter SBL [11] addresses sparse channel inference. Accordingly, neither processing AFDM in time nor combining SP with sparse Bayesian estimation is the contribution. The distinction is an executable *finite-list computation–fidelity contract*: retain the original full fractional-path likelihood, stop provably negligible candidate evaluations, and quantify the soft/channel information thereby omitted.

B. Contributions and scope

The paper makes three connected contributions. First, it formulates full fractional-Doppler SP-AFDM list evaluation

using inexpensive exact time-domain path rows and a properly normalized monotone marginal metric. Second, it develops a probability-reserve controller that preserves finite-list MAP and bounds posterior, bit-probability, and channel-mean distortion. Third, it evaluates the resulting cost–fidelity tradeoff against complete scoring, hard pruning, fixed-margin stopping, and clipped max-log counterhypothesis pruning, with separate estimated-geometry and frozen-receiver interface tests.

The result is a receiver component, not a replacement for path acquisition, OSD, or the entire turbo receiver. The Gaussian integration and recursion are classical; the claimed contribution is the coupled AFDM/SP implementation, soft-fidelity controller, and its quantified operating envelope. No global-codebook coverage or universal complexity advantage is asserted.

II. System Model and Finite-List Objective

A. Full fractional-Doppler SP-AFDM

An information message maps to a binary codeword $c \in \mathcal{C} \subseteq \{0,1\}^{2N}$ and a data vector $d(c) \in \mathbb{C}^N$. The transmitted transform-domain vector is

$$x = p + d(c), \qquad y = Hx + w, \quad w \sim \mathcal{CN}(0, \sigma^2 I_N), \tag{1}$$

where the deterministic pilot p and the data already include power scaling. The convention is $\mathbb{E}|w_n|^2 = \sigma^2$. The experiments use $|p_n|^2 = \rho$ and QPSK data energy $P_d = 1-\rho$. Their cross term need not vanish in a particular block; no per-block renormalization is imposed.

Define

$$F_{mn} = N^{-1/2} e^{-j2\pi mn/N}, \quad \Lambda_c = \operatorname{diag}(e^{j2\pi c n^2}), \qquad A = \Lambda_{c_2} F \Lambda_{c_1}. \tag{2}$$

All indices are zero-based. For integer delay l and real normalized Doppler ν, let

$$U(l,\nu) = A C_l \Delta_\nu \Pi_l A^{\mathsf{H}}, \qquad H = \sum_{s=1}^{S_0} h_s U(l_s, \nu_s), \tag{3}$$

where $[\Pi_l X]_n = X_{(n-l) \bmod N}$, $[\Delta_\nu]_{nn} = e^{j2\pi\nu n/N}$, and

$$[C_l]_{nn} = \chi_l(n) = \begin{cases} e^{j2\pi c_1[N^2 + 2N(n-l)]}, & n < l, \\ 1, & n \geq l. \end{cases} \tag{4}$$

This positive-chirp/positive-Doppler convention is used throughout and matches the evaluated implementation. Integer sample delays, a sufficient chirp-periodic prefix (CPP), ideal synchronization, and blockwise-constant path parameters are assumed. Doppler is not rounded in physical propagation. Fractional delay, unknown pulse shaping, and inter-block drift require additional modeling.

Each path operator is unitary, but the multipath sum generally is not. Full fractional propagation is used even when the front-end receiver adopts a restricted integer dictionary. The supplement derives the complete integer expansion and explains why deleting its tails is a model approximation rather than a harmless change of coordinates.

B. Frozen scoring model

For one evaluator call, freeze a set of S working path hypotheses $\{(\widehat{l}_s, \widehat{\nu}_s)\}_{s=1}^{S}$, a noise variance, and an exogenous prior

$$g \sim \mathcal{CN}(0, C_g), \qquad C_g = C_g^{\mathsf{H}} \succ 0. \tag{5}$$

The working geometry need not equal the true geometry in (3), and S need not equal S_0. Hypotheses may be supplied by an earlier estimator. The principal matched tests use known geometry and unknown gains; the estimated-geometry tests do not supply true delays, Dopplers, or gains to the scorer.

The zero-mean prior is not an estimated channel mean recycled as independent evidence from the same observation. Bayesian interpretation requires the stated generative assumptions. Under estimated geometry or other mismatch, the following guarantees remain pointwise comparisons between two evaluations of the same frozen working likelihood. They do not certify that likelihood as the true posterior.

C. Candidate list and outputs

A front end produces an ordered, duplicate-free list

$$\mathcal{B}(y) = \{d_1, \ldots, d_B\}. \tag{6}$$

OSD is used in the numerical implementation, but the theory does not require a particular proposal. The list may depend on y and is frozen before evaluation. Equal message priors are used in all tests. Decoder APPs influence proposal ordering only; they are not multiplied into the physical likelihood as additional independent measurements.

Let $\ell_N(d)$ be the complete normalized marginal cost derived below. The reference distribution is

$$P_B(d \mid y) = \frac{e^{-\ell_N(d)}}{\sum_{e \in \mathcal{B}} e^{-\ell_N(e)}}, \quad d \in \mathcal{B}. \tag{7}$$

The objective is to reduce evaluation work while retaining the reference MAP word and ensuring a specified bound on the distributional error introduced *within* $\mathcal{B}$. The principal soft output is the posterior gain mean; bit-probability intervals are also available. A guaranteed approximation to an arbitrary outer-loop CE algorithm is not implied.

III. Proposed Certified Marginalized Evaluator

A. Exact candidate regressors

For a complete coded candidate, compute

$$Y = A^{\mathsf{H}} y, \qquad X_d = A^{\mathsf{H}}(p + d). \tag{8}$$

The time-domain observation model is $Y_n = f_{d,n} g + W_n$, with row entries

$$f_{d,n,s} = \chi_{\widehat{l}_s}(n) e^{j2\pi\widehat{\nu}_s n/N} X_d[(n - \widehat{l}_s) \bmod N]. \tag{9}$$

The rows form $\Phi_d \in \mathbb{C}^{N \times S}$. Unitarity preserves the likelihood and thermal-noise variance. One IDAFT per candidate followed by shifted reads and phase products implements (9); a dense fractional-Doppler matrix is unnecessary.

Importantly, no independent-QPSK prior is assigned to time-domain samples. A complete valid transform-domain

codeword is generated first. The likelihood factorization conditions on that entire word. In general, a partial AFDM data-symbol sequence does not determine a corresponding independent observation prefix; this prevents directly treating the algorithm as a subcarrier-symbol tree search.

B. Complete marginalized likelihood

Under (5),

$$Y \mid d \sim \mathcal{CN}(0, C_d), \qquad C_d = \sigma^2 I_N + \Phi_d C_g \Phi_d^{\mathsf{H}}. \tag{10}$$

Subtracting the common noise-density normalization gives

$$\ell_N(d) = Y^{\mathsf{H}} C_d^{-1} Y + \log\det(I_N + \sigma^{-2}\Phi_d C_g \Phi_d^{\mathsf{H}}). \tag{11}$$

Equivalently, define

$$Q_d = C_g^{-1} + \sigma^{-2}\Phi_d^{\mathsf{H}}\Phi_d, \qquad b_d = \sigma^{-2}\Phi_d^{\mathsf{H}} Y. \tag{12}$$

Then

$$\ell_N(d) = \frac{\|Y\|^2}{\sigma^2} - b_d^{\mathsf{H}} Q_d^{-1} b_d + \log\det Q_d + \log\det C_g, \tag{13}$$

$$m_d = \mathbb{E}[g \mid d, Y] = Q_d^{-1} b_d, \quad V_d = Q_d^{-1}. \tag{14}$$

The determinant term is candidate dependent. Removing it gives a regularized profile criterion, not a cheaper evaluation of the same marginal likelihood. Complete gain-domain batch evaluation of (13) is therefore an essential same-target baseline.

C. Observation-prefix evaluation

Choose a row permutation π, fixed during the call. Descending $|Y_n|^2$ is the tested heuristic; it is not assumed optimal. Initialize $m_0 = 0$, $V_0 = C_g$, and $\ell_0 = 0$. At row $f = f_{d,\pi(t)}$, set

$$a_t = V_{t-1} f^{\mathsf{H}}, \quad v_t = \sigma^2 + f a_t,$$

$$e_t = Y_{\pi(t)} - f m_{t-1}, \tag{15}$$

$$\ell_t = \ell_{t-1} + |e_t|^2 / v_t + \log(v_t/\sigma^2), \tag{16}$$

$$m_t = m_{t-1} + a_t e_t / v_t,$$

$$V_t = V_{t-1} - a_t a_t^{\mathsf{H}} / v_t. \tag{17}$$

Theorem 1 (Exact monotone prefix)**.** *In exact arithmetic, $V_t \succeq 0$, $v_t \geq \sigma^2$, and ℓ_t equals the normalized marginal cost for the first t selected observations. In particular,*

$$0 \leq \ell_t \leq \ell_{t+1} \leq \ell_N. \tag{18}$$

Proof. Gaussian conditioning gives (15) and (17). Multiplication of the predictive densities factors the joint marginal density. Subtracting $t \log \sigma^2$ from its log-determinant contribution gives (16). Both incremental terms are nonnegative because $V_{t-1} \succeq 0$. At $t = N$ this factorization equals (11). □

The normalization $\log(v_t/\sigma^2)$ is necessary. Using $\log v_t$ can yield negative increments when $v_t < 1$. An observation-dependent permutation does not change the algebra for the realized arrays: the prefix is a fixed-subset density, not a density additionally conditioned on the row-selection event.

Maintain the best fully evaluated cost ℓ_*. A candidate with $\ell_t > \ell_*$ cannot win. Strict comparison and retaining the first minimizer preserve ties. Nevertheless, hard pruning alone may remove a substantial sum of nonwinning weights. The next subsection controls this distinct loss.

Algorithm 1 Certified soft-output finite-list evaluation

Require: Frozen y, p, path hypotheses, $C_g \succ 0$, $\sigma^2 > 0$, list $\mathcal{B}$, tolerance ϵ

Compute Y, all X_d, path phases, and row order π
$Z_K \leftarrow 0$; $U_P \leftarrow 0$; $\ell_* \leftarrow +\infty$; $\mathcal{K}, \mathcal{P} \leftarrow \varnothing$
for d in the declared list order **do**
  Set $m \leftarrow 0$, $V \leftarrow C_g$, $\ell \leftarrow 0$
  Count remaining words R; compute a_d by (20)
  Set threshold to $+\infty$ for the first word or $a_d \leq 0$; otherwise use (21)
  for $t = 1, \ldots, N$ **do**
    Form row (9); update (15)–(17)
    if $t < N$ and the stopping threshold is crossed **then**
      Store prefix cost; add d to $\mathcal{P}$; stop this word
    end if
  end for
  if all observations were processed **then**
    Store m_d and exact weight; add d to $\mathcal{K}$
    Update incumbent using strict improvement
  end if
  Update Z_K or U_P using log-sum-exp
end for
return list-MAP word, completed-word weights, channel mean, bit intervals, mass/mean bounds, work count

D. Adaptive probability reserve

Let $\mathcal{K}$ contain completed candidates and $\mathcal{P}$ contain stopped candidates. Store the final partial cost $\underline{\ell}_d$ of each stopped candidate, and define

$$Z_K = \sum_{d \in \mathcal{K}} e^{-\ell_N(d)}, \qquad U_P = \sum_{d \in \mathcal{P}} e^{-\underline{\ell}_d}. \tag{19}$$

For a requested $0 < \epsilon < 1$, put $r = \epsilon/(1-\epsilon)$. Complete the first word. Before a subsequent word, let R be the number of words still to assess, including the current word, and allocate

$$a_d = \frac{r Z_K - U_P}{R}. \tag{20}$$

When $a_d \leq 0$, fully evaluate the candidate. Otherwise stop it only when

$$\ell_t > \max\{\ell_*, -\log a_d\}. \tag{21}$$

A completed candidate adds its exact weight to Z_K; a stopped candidate adds its prefix upper weight to U_P. The incumbent is updated only by completed candidates. Overshoot below a candidate's allowance remains available for subsequent words. This equal-remaining allocation is inexpensive, not claimed to be complexity optimal.

A simpler uniform-reserve comparator uses the same recursion and the threshold

$$\ell_t > \ell_* + T_\epsilon, \qquad T_\epsilon = \max\left\{0, \log\frac{B(1-\epsilon)}{\epsilon}\right\}. \tag{22}$$

Both rules form one certified-evaluation method family. The adaptive-versus-uniform difference is an ablation, not a separate contribution.

IV. Finite-List Fidelity Guarantees

A. MAP and posterior-mass preservation

Theorem 2 (Certified reserve)**.** *Upon completing Algorithm 1, its hard decision equals the MAP decision in $\mathcal{B}$. If $\epsilon_B = P_B(\mathcal{P} \mid y)$, then*

$$\epsilon_B \le \bar{\epsilon}_B \triangleq \frac{U_P}{Z_K + U_P} \le \epsilon. \tag{23}$$

The same conclusion holds for (22)*.*

Proof. Theorem 1 makes $e^{-\underline{\ell}_d}$ an upper bound on each stopped word's complete weight. Hence the unknown discarded weight Z_P obeys $Z_P \le U_P$, and $Z_P/(Z_K + Z_P) \le U_P/(Z_K + U_P)$. Initially $U_P = 0$. Completion increases Z_K without increasing U_P. Rejection adds at most $(rZ_K - U_P)/R \le rZ_K - U_P$ to U_P. Induction proves $U_P \le rZ_K$, which yields the last inequality in (23). The independent incumbent condition in (21) ensures that a potentially winning word is never rejected. For the uniform rule, every stopped upper weight is at most $e^{-\ell_*^{\mathrm{final}} - T_\epsilon}$, since the incumbent cost never increases. At most B words are stopped, while $Z_K \ge e^{-\ell_*^{\mathrm{final}}}$. Thus $U_P/Z_K \le Be^{-T_\epsilon} \le r$. The case $T_\epsilon = 0$ satisfies $B \le r$ by its definition. MAP preservation follows because $T_\epsilon \ge 0$. □

The theorem bounds a sum under the actual scoring model. It is neither a Monte Carlo confidence interval nor a proxy decoder's confidence approximation. Only completed candidates supply full-observation conditional means; intermediate m_t from stopped words must not be used as if they were m_d.

B. Soft probability and LLR outputs

Let $P_K = P_B(\cdot \mid \mathcal{K})$, extended by zero on stopped words. Then

$$\mathrm{TV}(P_B, P_K) = \epsilon_B \le \bar{\epsilon}_B. \tag{24}$$

On $\mathcal{K}$, $P_B = (1 - \epsilon_B)P_K$; direct summation of absolute differences proves (24). The same identity holds for the joint (d, g) mixture when each retained candidate keeps its exact conditional gain distribution. Any bit probability therefore changes by at most $\bar{\epsilon}_B$.

A global finite LLR-error bound does not follow from total variation alone. The likelihood ratio can be arbitrarily sensitive near a zero bit probability. Instead, the stopped words' known bit labels give explicit intervals. For one code bit, define completed weights Z_0, Z_1 and stopped upper weights U_0, U_1, so $Z_K = Z_0 + Z_1$ and $U_P = U_0 + U_1$.

Corollary 1 (Bit-specific intervals)**.** *The full-list probability $q = P_B(c_i = 1 \mid y)$ satisfies*

$$\frac{Z_1}{Z_K + U_0} \le q \le \frac{Z_1 + U_1}{Z_K + U_1}. \tag{25}$$

With $L = \log[P_B(c_i = 0 \mid y)/P_B(c_i = 1 \mid y)]$,

$$\log \frac{Z_0}{Z_1 + U_1} \le L \le \log \frac{Z_0 + U_0}{Z_1}, \tag{26}$$

using extended-real endpoints where needed.

Proof. Write the unknown stopped weights as $0 \le W_j \le U_j$. The ratio $(Z_1 + W_1)/(Z_K + W_0 + W_1)$ increases with W_1 and decreases with W_0. Its extrema yield (25); the odds ratio gives (26). □

These intervals require only accumulation of available candidate labels and weights. They do not estimate probability outside $\mathcal{B}$, and they are not a guarantee on extrinsic messages after an unspecified iterative receiver reuses them.

C. Posterior channel-mean control

The completed-candidate estimate is

$$\widehat{g}_K = \sum_{d \in \mathcal{K}} \frac{e^{-\ell_N(d)}}{Z_K} m_d, \qquad \widehat{g}_B = \mathbb{E}_{P_B}[g]. \tag{27}$$

Theorem 3 (Channel-mean fidelity)**.** *For*

$$R_g = \frac{\sqrt{\lambda_{\max}(C_g)}\, \|Y\|}{2\sigma}, \tag{28}$$

every complete conditional mean satisfies $\|m_d\| \le R_g$, and

$$\|\widehat{g}_B - \widehat{g}_K\| \le 2R_g \bar{\epsilon}_B. \tag{29}$$

Proof. Set $T_d = \Phi_d C_g^{1/2}/\sigma$. Equation (14) can be written

$$m_d = C_g^{1/2} T_d^{\mathsf{H}} (I + T_d T_d^{\mathsf{H}})^{-1} Y/\sigma. \tag{30}$$

The middle matrix has singular values $t/(1 + t^2) \le 1/2$, proving (28). The complete mean equals $(1 - \epsilon_B)\widehat{g}_K + \epsilon_B \widehat{g}_P$, where both conditional mixture means lie in the radius-R_g ball. Taking their difference proves (29). □

For an absolute approximation tolerance $\tau > 0$, use

$$\epsilon_{\mathrm{use}} = \min\{\epsilon_0, \tau/(2R_g)\}, \tag{31}$$

with the zero-observation case handled directly. This controls the change from the full-list posterior channel mean, not true-channel NMSE. It is also distinct from the frozen-distribution least-squares update in the supplement; those two estimators must not be silently identified.

D. Coverage and model error

Let P be the full-codebook posterior under the same working model, and $\epsilon_{\mathrm{gen}} = P(\mathcal{C} \setminus \mathcal{B} \mid y)$. The total excluded mass is

$$\epsilon_{\mathrm{gen}} + (1 - \epsilon_{\mathrm{gen}})\epsilon_B. \tag{32}$$

Only ϵ_B is controlled here. A larger list can reduce the first component, but no such reduction is certified without further information. Likewise, mismatch between the frozen geometry/prior and the physical channel is not removed by exact evaluation. The theorem separates computational approximation from these other errors rather than asserting their absence.

V. Complexity and Receiver Integration

Let $M = \sum_{d\in\mathcal{B}} t_d \le BN$ be the evaluated observation-row count. Precomputed path phases cost $O(NS)$; each evaluated row costs $O(S^2)$. For an implementation that stores all transformed candidates, the online cost is

$$C_{\text{eval}} = C_{\text{proposal}} + O(BN\log N) + O(MS^2) + O(BS+NS) + O(Bn_c), \tag{33}$$

where $n_c = 2N$ and the last term applies when all bit intervals are returned. Sorting the observations adds $O(N\log N)$. Checking/factoring the common prior is $O(S^3)$. Memory is $O(BN+BS+NS+S^2)$, apart from the front-end code representation. Candidate streaming can reduce storage, but that unimplemented saving is not credited in the measurements.

Unpruned sequential evaluation has $M = BN$ and uses the same arithmetic per row. Gain-domain batch marginalization additionally forms/factors a candidate information matrix. These are more informative implementation controls than comparison with a deliberately dense $N \times N$ inversion for every word. Worst-case complexity remains full evaluation. Diffuse likelihoods, large S, near-indistinguishable hypotheses, or weak front-end reliabilities can eliminate the saving.

The component attaches after a front end has produced an observation-dependent list and frozen geometry. Returned support identifiers, reliabilities and observations are sufficient; true data and true gains are not accepted. A terminal rescue call can return the list-MAP word and a gain-mean approximation. Feeding these outputs back into a changing turbo loop requires a separately specified interface and does not inherit multi-iteration MAP or FER guarantees from one call.

Complete latency obeys

$$T_{\text{total}} = T_{\text{front}} + T_{\text{proposal/transform}} + T_{\text{score/soft}}. \tag{34}$$

If a fraction α of baseline time is accelerated by a factor s, the complete speedup is $[1-\alpha+\alpha/s]^{-1}$. A reduction in row count therefore cannot be advertised as the same percentage of total receiver time. A hard fixed-row-budget variant, described only in the supplement, has a different contract and does not simultaneously guarantee fixed total latency and the soft tolerance.

The implementations use log-sum-exp weights, strict comparisons with a conservative numerical margin, and positive-variance checks. Theorems concern exact arithmetic; the tests are not formal interval arithmetic. A failed covariance check is a numerical failure, not evidence that a candidate has negligible mass. A production implementation should use a tested square-root or stable-factorization fallback and include its cost.

VI. Numerical Results

A. Experiment design and baselines

All scored observations use the full fractional-path model; no tail renormalization is applied. The controlled suite contains 192 independent channel/message/noise draws, paired across arms: $N \in \{48, 96, 192\}$, $S \in \{3, 6\}$, $\rho \in \{0.15, 0.5\}$, and $\sigma^2 \in \{0.05, 0.20\}$, with eight draws per cell. Gallager codes of length $2N$ use $d_v = 3$, $d_c = 6$ and seed 20260726. Lists contain 256 distinct candidates. The working covariance is proportional to $0.45^{i+j}0.35^{|i-j|}$. Six-path stress instances additionally normalize the true gains and perturb actual Dopplers by $0.06\sin(s+0.3)$ relative to the scorer. Such mismatch tests the approximation to the frozen model, not the correctness of that model.

TABLE I
Controlled suite: 192 paired draws, $B = 256$, $\epsilon = 0.01$

Quantity	Uniform	Adaptive
Mean row fraction	0.435906	0.432639
Maximum actual discarded mass	0.000144	0.002907
Maximum reported bound	0.004106	0.009922
Maximum gain-mean change	1.18×10^{-5}	9.20×10^{-5}
MAP disagreements	0/192	0/192

Complete batch and compiled sequential marginalization evaluate the same list likelihood. Hard pruning uses the incumbent only. Uniform and adaptive reserve use $\epsilon = 0.01$. A fixed-margin control uses the *same* marginal score and stops at $\ell_t > \ell_* + \log 100$; it preserves MAP but lacks a list-size-aware mass guarantee. This is a declared same-model heuristic, not a claimed faithful reproduction of a published sphere decoder or soft-output OSD receiver. A further counterhypothesis-pruning control preserves the exact clipped max-log bit metrics of the same list, with clipping level 5. It adapts the clipping objective to the marginal score rather than reproducing a published symbol-tree implementation; its stopping rule is specified in the supplement.

The uniform policy was fixed before the controlled suite. Adaptive reserve was subsequently evaluated on those same draws; its incremental advantage is an ablation. Both were fixed before the new estimated-geometry test below. The accompanying supplement records streams, filenames, and timing scopes. External papers' published curves are not represented as independently reproduced here.

B. Large-list fidelity and work reduction

Table I confirms exact hard-decision agreement and the prescribed mass/mean bounds. Adaptive reserve uses more of the permitted probability budget; its larger approximation error remains within that budget. Figure 1 shows how work varies with block length and noise. The gain over uniform reserve is modest and is not the principal performance claim.

The full 256-word list and both certified evaluators produce 48 failed frames. Fully evaluating only the first 16 or 64 nested candidates produces 64 and 54 failures. The decoding improvement belongs to the larger candidate set; certified pruning preserves its result rather than changing a fixed list's decision.

A complete 256-word codebook at $N = 12$ supplies a counterexample to hard-only soft output. Hard pruning preserves global codeword MAP, yet discards 0.748950 probability and changes the posterior gain mean by 0.289413. This selected witness illustrates a failure mechanism, not its average prevalence.

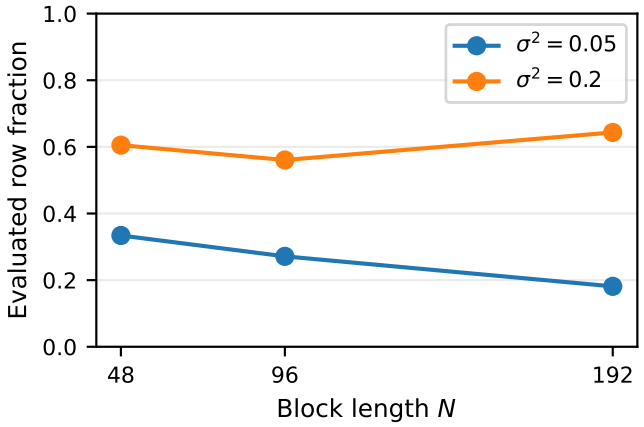


Fig. 1. Evaluated row fraction for the same 256-candidate list. Each noise curve averages 96 controlled draws over the declared path-count and pilot-power settings.

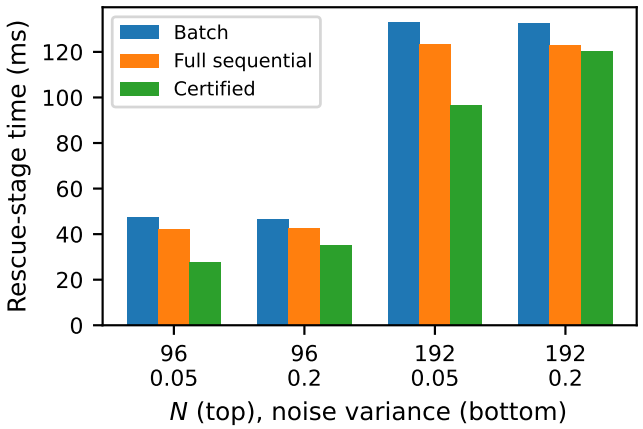


Fig. 2. Same-list rescue-stage timing, including proposals and candidate transforms but excluding prior CE/LLR generation. All three arms return the same MAP word.

TABLE II
RESCUE-STAGE MEDIANS IN MS, $B = 1024$; 15 INTERLEAVED REPETITIONS PER FIXTURE

N	σ^2	Batch	Full sequential	Certified
96	0.05	47.400	42.170	27.519
96	0.20	46.752	42.474	34.974
192	0.05	132.953	123.552	96.500
192	0.20	132.783	122.875	120.237

TABLE III
ESTIMATED-GEOMETRY CHECK: SAME-SCORE PRUNING CONTROLS, 32 PAIRED DRAWS

Rule	Row fraction	Max. mass	$> 1\%$
Hard	0.6832	0.675320	15/32
Margin log 100	0.7128	0.053959	4/32
Clipped max-log	0.7137	0.140294	3/32
Uniform reserve	0.7444	0.000215	0/32
Adaptive reserve	0.7382	0.002186	0/32

C. Equal-power and high-uncertainty boundary

A separate 48-draw test uses $C_g = I/S$, $N = 48, 96$, $S = 3, 6$, $\rho = 0.5$, and $\sigma^2 = 0.05, 0.2, 0.5$. All decision, mass, and mean-bound checks pass. For six paths the uniform policy evaluates 29.16%, 74.81%, and 93.87% of rows as noise increases. Imposing $\tau = 10^{-3}$ via (31) raises these to 29.94%, 77.37%, and 96.72%. Thus fidelity persists in a diffuse regime in which almost no acceleration remains. A guaranteed average saving is not inferred from favorable low-noise cases.

D. Representative rescue-stage timings

Every arm in Table II includes OSD from existing LLRs, QPSK mapping, candidate IDAFTs, scoring, and the posterior gain mean. Initial CE and the equalizer/LDPC processing that produced the LLRs, common observation transforms, code construction, and JIT compilation are excluded. The same compiled recursion is used with and without pruning.

At $\sigma^2 = 0.05$, time reductions against that unpruned recursion are 34.74% at $N = 96$ and 21.90% at $N = 192$. At $N = 192$, $\sigma^2 = 0.20$, the reduction is 2.15%. Figure 2 retains this unfavorable point. These are fixture-specific medians, not hardware-independent confidence statements or complete-receiver percentages.

E. Observation-estimated geometry and soft controls

A new 32-draw test freezes both certified policies and estimates geometry from the observation. The true channel has three full fractional paths, distinct delays selected from $\{0, \ldots, 4\}$, and Dopplers uniformly distributed on $[-1.4, 1.4]$. A pilot-only OMP front end selects three atoms from delays 0–4 and Dopplers $-1.5 : 0.25 : 1.5$, then refits their gains. LMMSE reliabilities generate the 256-word OSD list. The scorer uses all three estimated atoms, $C_g = I/3$, and the known thermal variance. No actual path geometry or gains enter this scoring call. Four fresh draws are used per (N, ρ, σ^2) cell for $N = 48, 96$ and the same two powers/noise levels.

Every decision agrees with complete scoring. However, the fixed margin violates the requested 1% mass tolerance in four draws, while hard-only pruning violates it in 15. The clipped max-log control also preserves all clipped bit metrics, with maximum discrepancy 2.28×10^{-13}, but discards more than 1% posterior mass in three draws. Its maximum channel-mean change is 0.004409. This does not invalidate max-log decoding: it demonstrates that preserving clipped minima is a different contract from preserving posterior sums. The certified rules satisfy their bounds in every draw. Maximum bit-probability changes are 0.5453 for hard pruning, 0.02064 for the fixed margin, and 0.001317 for adaptive reserve. The clipped control was added after the estimated-geometry test and is a post-synthesis ablation; no dedicated timing advantage is claimed for it. All new bit intervals contain the complete-list bit probabilities. Figure 3 separates lower work from the fidelity requirement.

The cold-start front end is deliberately not a tuned production receiver: complete-list decoding fails on 27/32 draws. This high error rate must not be interpreted as a benefit over a strong

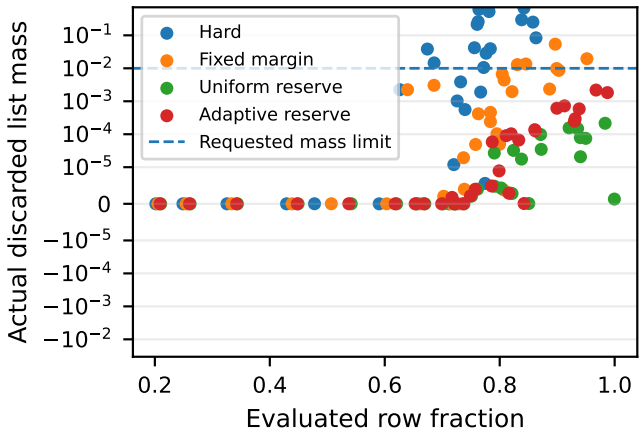


Fig. 3. New observation-estimated-geometry check. Each point is one of the four displayed rules on one draw; the additional clipped max-log comparison is in Table III. The reference line is the requested 1% mass tolerance; the vertical axis is symmetric-logarithmic near zero.

turbo baseline. It establishes that the fidelity contract survives observation-dependent geometry, not that such geometry is adequate. Including pilot fitting, LMMSE, OSD, transforms, and soft output, mean measured stage sums are approximately 12.31 versus 11.85 ms at $N = 48$, and 28.36 versus 27.23 ms at $N = 96$, for full versus adaptive evaluation. Static code/dictionary construction and the common final row-order sort are outside these new timers. These small reference-implementation reductions are less than the row-count reductions.

F. Frozen-receiver attachment and complete cost

One completed engineering block attaches the evaluator to the supplied formal I1 adapter at $N = 1024$, using its frozen checkpoint, the $(2048, 1024)$ code, recipe B0-p2-s0.5, a legal decoder-feedback checkpoint, and 12 outer BP iterations per scheduled detection. The physical block contains eight full fractional paths. The scorer receives the resulting 24 estimated integer support atoms and terminal LLRs; it receives no true support. It uses the exogenous $I/24$ prior and the same known thermal variance for all scoring arms. Neither the frozen receiver nor its feedback policy is modified.

The 64-candidate full, uniform, and adaptive scorers agree on MAP and all fidelity checks. Adaptive evaluation uses 92.87% of the full row work; the actual discarded mass is 5.72×10^{-17} and its reported upper bound is 0.009598. The transmitted word is absent from the list, and both the front end and rescue decision fail. This single block is an interface check, not a FER estimate.

Measured stage times are 3.480 s for the front end, 9.260 s for proposal/transforms, and 0.186 versus 0.168 s for full versus adaptive scoring/soft output. Thus the summed reference times are 12.926 versus 12.908 s: only about 0.14% improvement, which is not statistically established. Common support-to-regressor preparation and row ordering are outside these stage timers, so the sums are not stop-to-stop latency measurements. The dominant measured cost is this generic large-code OSD implementation. This result prevents translating the moderate-dimensional rescue-stage speedups into a complete-I1 claim. No rescue-output feedback into the frozen loop is exercised.

G. Independent verification

The original controlled suite is checked through separately constructed observation-domain covariance solves; 3,392 comparisons have maximum discrepancy 4.37×10^{-11}. The adaptive bookkeeping check adds 1,536 comparisons. A fresh independent program for the present estimated-geometry and frozen-interface tests imports no candidate algorithm. It uses dense physical definitions and, at $N = 1024$, augmented least squares to recompute marginalized costs and means. Its 627 comparisons pass, with maximum discrepancy 1.35×10^{-11}. A separate 72-comparison check verifies retained physical/statistical identities and the clipped max-log control, with maximum discrepancy 9.10×10^{-13}. These are numerical consistency checks, not numbers of independent scientific replications. Exact-arithmetic proofs establish the stated identities; floating-point comparisons check their implementations.

VII. Limitations and Scope

The selected operating regime is finite-list rescue or short-packet joint reception with moderate gain dimension and enough likelihood separation to stop many candidates early. Three boundaries remain material. First, the method neither generates missing codewords nor certifies their probability. Second, geometry acquisition and gain-prior calibration are separate statistical problems; the same-model certificate survives mismatch but does not compensate for it. Third, proposal generation and front-end processing may dominate the complete cost, as the full-size attachment demonstrates.

The experiments establish a component-level fidelity–cost advantage and an executable terminal interface. They do not establish superiority over the complete frozen I1 receiver, effective production-scale fractional-geometry estimation, or beneficial feedback of every approximate soft output. Neither the fixed-margin nor the counterhypothesis control is described as a faithful implementation of every published soft sphere/list algorithm. The latter is tested against its own exact clipped max-log target. A complete-receiver comparison must additionally freeze the surrounding interface, include all calls and conversions, compare against appropriate same-objective and established soft-search controls, and use fresh radio blocks. The companion package supplies an executable independent-confirmation configuration; a large production confirmation is not claimed here.

VIII. Conclusion

A certified evaluator has been developed for finite codeword lists in full fractional-Doppler coded SP-AFDM. Exact time-domain regressors and normalized Gaussian innovations enable early termination of candidate computations, while a probability reserve preserves list MAP and controls

posterior/channel-mean distortion. Controlled experiments establish a scoped reduction in likelihood work and rescue-stage time. Estimated-geometry and frozen-receiver tests verify the interface and expose the effects of model mismatch and proposal cost. The resulting contribution is a finite-list computation–fidelity tradeoff, not a guarantee of global-codebook coverage or complete-receiver acceleration.

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